

```
# find which reference category matches stata's for model 7
for (ref in all_depts) {
  dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
  f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
                             ref_partic + share_santos_2_100 +
                             incidence_multidim_pov_2005_100 + lpop + indrural2005 +
                             cultivated100 + oil + ln_altitud +
                             coberturabrutaeduc +", dept_vars_try))
  model_try <- lm(f_try, data = df)
  intercept <- coef(model_try)["(Intercept)"]
  if (abs(intercept - 18.78) < 0.01) {
    cat("Model 7 reference category:", ref, "\n")
  }
}
```